The RSA-2048 Reality Check: A Surface Code Analysis

BreaQ Hackathon: Blue Track Submission

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1 Current Estimates and the Qubit-Runtime Tradeoff

The landscape of quantum cryptanalysis is defined by extreme tradeoffs between physical qubit counts and execution time. In 2019, Gidney and Ekerå estimated that factoring RSA-2048 would require 20 million physical qubits running for 8 hours. By 2025, Gidney revised this estimate down to 1 million physical qubits, but at the cost of extending the runtime to roughly one week.

This 20x reduction in physical qubits—without any assumed changes in hardware capabilities (0.1% gate error, 1 μs cycle time)—is purely algorithmic. It relies on techniques like yoked surface codes, approximate residue arithmetic, and aggressive magic state cultivation. By packing logical qubits more densely and performing fewer parallel operations, the algorithm trades space (qubits) for time (circuit depth).

Recent 2026 estimates continue this trend. Iceberg Quantum proposed breaking RSA-2048 with under 100,000 qubits using Quantum Low-Density Parity-Check (QLDPC) codes, though this relies on long-range connectivity not yet realized in hardware. Meanwhile, Google Quantum AI estimated breaking ECC-256 in under 9 minutes with 500,000 qubits, utilizing zero-knowledge proofs to verify their withheld circuit models.

2 Surface Code Error Estimation

To understand the engineering reality, we construct a toy surface-code model to factor RSA-2048. We assume a requirement of $L = 1,500$ logical qubits running for 5 days at a 1μs cycle time, yielding $T \approx 4.3 \times 10^{11}$ total cycles. The protocol must complete with a failure probability of less than 10%, meaning $L \cdot T \cdot p_L < 0.1$.

Solving for the maximum allowable logical error rate per cycle (p_L):

$$p_L < \frac{0.1}{1,500 \cdot 4.3 \times 10^{11}} \approx 1.55 \times 10^{-16}$$

Assuming a physical gate error rate of $p = 0.001$ (0.1%) and a threshold of $p_{th} = 0.01$ (1%), we determine the required surface code distance d using the empirical scaling law:

$$p_L \approx 0.1 \left(\frac{p}{p_{th}} \right)^{\frac{d+1}{2}} = 0.1(0.1)^{\frac{d+1}{2}}$$

Setting $p_L = 1.0 \times 10^{-16}$ to satisfy our bound, we find:

$$10^{-16} = 10^{-1} \cdot 10^{-\frac{d+1}{2}} \implies 15 = \frac{d+1}{2} \implies d = 29$$

With a code distance of $d = 29$, each logical qubit requires approximately $2d^2 = 2(29)^2 = 1,682$ physical qubits. Multiplying this by our $1,500$ logical qubits yields a baseline requirement of approximately **2.52 million physical qubits**, perfectly aligning with the order of magnitude in modern estimates before accounting for magic-state distillation factories.

3 Hardware Reality and The Post-Quantum Transition

Today's most advanced quantum processors (e.g., IBM Condor) feature approximately $1,121$ physical qubits with two-qubit error rates hovering around 0.5%. Comparing our calculated requirement of 2.5 million qubits to a 1,000-qubit processor reveals a **2,500x gap** in scale, alongside a critical gap in error rates (requiring a $5x$ reduction to reach 0.1%).

Despite this massive hardware gap, the cryptographic timeline requires immediate action due to the "Harvest Now, Decrypt Later" (HNDL) threat model. Adversaries are currently intercepting and storing encrypted network traffic. If hardware scales at the pace required to close the 2,500x gap within 15 years, data captured today will be exposed. Therefore, organizations must immediately begin migrating to post-quantum algorithms like ML-KEM (Kyber) for key encapsulation to secure current communications against future quantum decryption.